Ultra-Low-Power ESP32 Smartwatch
Project Proposal

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1 Application / Problem Statement

With the rapid growth of wearable technology, smartwatches have become an essential tool for timekeeping, fitness tracking, and connectivity. However, most commercially available smartwatches suffer from a critical limitation: **poor battery life**. Frequent charging (often daily) significantly reduces usability and user convenience.

Additionally, many smartwatches rely on power-hungry displays and continuously active wireless modules, making them inefficient for basic use cases such as timekeeping, alarms, and lightweight health monitoring.

There is a clear need for a **low-power, cost-effective smartwatch** that:

- Provides essential features like time display, stopwatch, and notifications
- Maximizes battery life (multi-day operation)
- Maintains readability in all lighting conditions
- Allows modular expansion for health monitoring

2 Proposed Solution

To address the above challenges, we propose the design and development of an **Ultra-Low-Power Smartwatch** based on the ESP32-S3 microcontroller.

Core Architecture

The system is built around the **ESP32-S3 dual-core microcontroller**, which provides integrated Wi-Fi and Bluetooth Low Energy (BLE) capabilities for future wireless synchronization and updates.

Display System

A **1.54-inch E-Ink (GDEY0154D67) display** with 200×200 resolution is used. Unlike traditional LCD/OLED displays, E-Ink:

- Consumes power only during refresh cycles
- Offers good sunlight readability
- Enables an always-on display with minimal power usage

SPI Communication Interface

The communication between the ESP32-S3 and the E-Ink display is implemented using the **Serial Peripheral Interface (SPI)** protocol. The ESP32 acts as the master, while the display functions as the slave device.

SPI enables fast and efficient transfer of command and display data using signals such as MOSI, SCLK, CS, and DC. Additional control lines like RST and BUSY ensure proper synchronization during display updates.

This interface is chosen for its high speed, low pin requirement, and suitability for display communication.

User Interface

The smartwatch uses a **non-touch physical interface** consisting of four push buttons for navigation. A **vibration motor** is integrated to provide haptic feedback for alerts and user interactions.

Power Management System

- Powered by a **3.7V Li-Po battery (200–500 mAh)**
- Uses a **low-dropout (LDO) regulator (NCP167AMX330TBG)** to generate stable 3.3V and 1.8V rails
- Includes a **battery charging IC** with constant-current (CC) and constant-voltage (CV) charging
- Protection circuitry ensures safety against over-voltage, short-circuits, and deep discharge

Ultra-Low-Power Optimization

The ESP32 operates in **deep sleep mode**, reducing standby current to approximately 10–150 μA. The system wakes up only when necessary using:

- Button interrupts
- Motion detection via accelerometer

Sensor Integration

- **BMA423 Accelerometer**: Used for motion detection, step counting, and wake-up triggering

Future Expansion

The I2C architecture allows easy integration of sensors like:

- **MAX30102**: Heart rate and SpO_2 monitoring

Debugging and Connectivity

A **CP2102N/CP2104 USB-UART bridge** is used for:

- Firmware uploading
- Serial debugging

An **ESD protection circuit** is included to safeguard the system from electrostatic discharge on USB lines.

3 Practical Feasibility and Cost Effectiveness

Component Availability

All components used in the design are:

- Commercially available through distributors such as Digi-Key, Robu, Amazon.
- Widely used in embedded and IoT applications
- Supported by strong documentation and community resources

Key components such as ESP32-S3, E-Ink display modules, Li-Po batteries, and sensor ICs are readily accessible, ensuring smooth procurement and development.

Cost Effectiveness

The design is optimized for affordability while maintaining performance. Key cost considerations:

- ESP32-S3 provides multiple functionalities (MCU + wireless), reducing component count
- E-Ink display reduces long-term energy cost and battery size requirements
- Use of standard passive components and widely available ICs minimizes BOM cost

Based on the provided Bill of Materials (BOM), the estimated prototype cost remains within a reasonable budget for academic and small-scale production purposes.

Power Efficiency Advantage

The major advantage of this design is its **extended battery life of 5–7 days**, achieved through:

- Low-power display technology
- Deep sleep optimization
- Event-driven system wake-up

This makes the device significantly more efficient compared to conventional smartwatches.

Implementation Feasibility

The project is highly feasible due to:

- Modular design architecture
- Availability of development tools for ESP32
- Simple PCB design with standard interfaces (I2C, UART, GPIO)

The system can be implemented, tested, and iteratively improved within a typical academic project timeline.
Total Component Cost : Rs.3,000
PCB Fabrication and Assembly : Rs.7,000
Total price : Rs.10,000

4 Conclusion

The proposed ultra-low-power smartwatch provides a practical and efficient solution to the limitations of existing wearable devices. By combining energy-efficient hardware, optimized power management, and scalable architecture, the design achieves a balance between functionality, cost, and battery life.